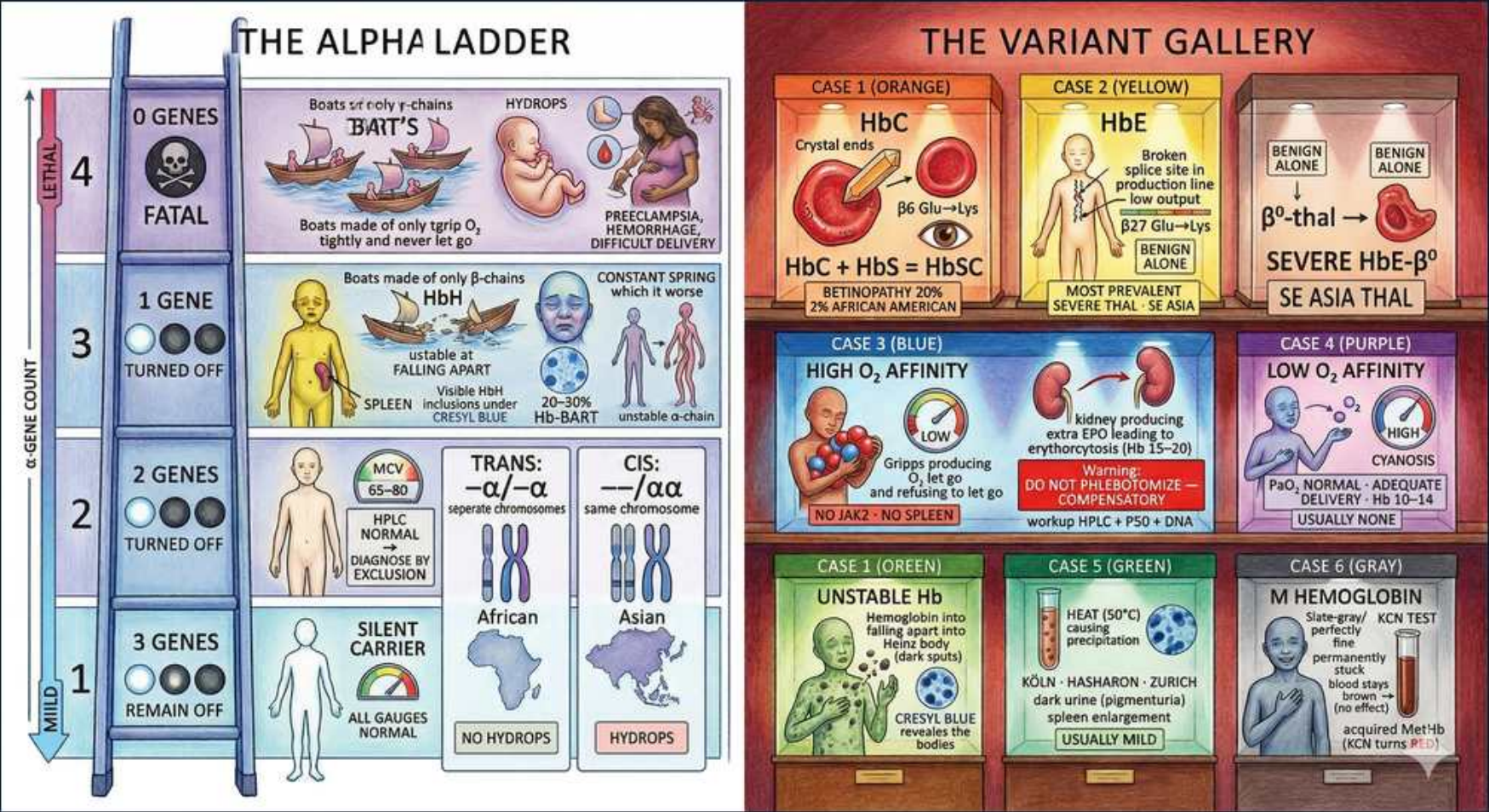


Hemoglobinopathy — The Alpha Ladder & Variant Gallery



1. 4 genes deleted — Bart/hydrops

WHAT YOU SEE

Ladder top: 0 genes, FATAL, Hb Barts, hydrops fetalis

WHAT IT ENCODES

There are 4 α -globin genes (two on each chromosome 16). Deletion of ALL 4 ($--/--$) leaves no α chains, so fetal γ chains self-associate into Hb Barts (γ_4), which has extremely high O2 affinity and cannot release O2 to tissues. Result: severe in-utero hypoxia $\rightarrow$ hydrops fetalis (high-output failure, anasarca), death in utero or shortly after birth. The mother is at risk of severe preeclampsia / mirror syndrome and difficult/hemorrhagic delivery.

BOARD TRAP

WRONG: 'Hb Barts (γ_4) carries oxygen well because it has four heme groups.' Its O2 affinity is so high it never unloads O2 — tissues are profoundly hypoxic, hence fatal hydrops.

2. 3 genes deleted — HbH

WHAT YOU SEE

1 gene left, HbH disease, β_4 tetramers, splenomegaly

WHAT IT ENCODES

Deletion of 3 of 4 α genes ($--/-\alpha$) leaves one functioning gene $\rightarrow$ marked α -chain deficit. Excess β chains form HbH (β_4), an unstable tetramer that precipitates into Heinz-like inclusions, causing a moderate-to-severe chronic hemolytic anemia with splenomegaly, jaundice, and microcytosis. HbH (5–30%) is detectable on hemoglobin electrophoresis/HPLC and with supravital (brilliant cresyl blue) staining showing a golf-ball/pitted inclusion pattern. Like HbH, Hb Barts also fails to deliver O2.

BOARD TRAP

WRONG: 'HbH disease is asymptomatic like α -thal trait.' Three-gene deletion produces a clinically significant transfusion-occasional hemolytic anemia with splenomegaly — not a silent state.

3. 2 genes — α -thal trait

WHAT YOU SEE

2 genes turned off, MCV 65-80, HPLC normal

WHAT IT ENCODES

Loss of 2 α genes = α -thalassemia trait (minor): mild or absent anemia with disproportionate microcytosis (MCV ~65–80) and a normal-to-high RBC count (Mentzer index <13). Crucially, electrophoresis/HPLC is NORMAL — HbA2 and HbF are NOT elevated — so α -thal trait is a diagnosis of exclusion (normal iron studies, normal HbA2) confirmed by genetic testing.

BOARD TRAP

WRONG: ' α -thal trait shows an elevated HbA2.' HbA2 is normal in α -thal trait; elevated HbA2 (>3.5%) is the signature of β -thalassemia trait — that mix-up is the classic board trap.

4. Cis vs trans / ethnicity

WHAT YOU SEE

trans ($-\alpha/-\alpha$ African) vs cis ($--/aa$ Asian)

WHAT IT ENCODES

The two missing genes can be in cis (both on the SAME chromosome, $--/aa$ — common in Southeast Asians) or trans (one on EACH chromosome, $-\alpha/-\alpha$ — common in those of African descent). This matters for offspring: a cis carrier can pass — to a child, so cis $\times$ cis risks HbH disease or Hb Barts/hydrops. Trans carriers cannot produce a — chromosome, so their offspring are not at risk for the severe forms — hence genetic counseling hinges on ethnicity/configuration.

BOARD TRAP

WRONG: 'Any couple who are both α -thal trait can have a hydrops fetalis baby.' Only the cis ($--/aa$, Asian) configuration can transmit a —; trans ($-\alpha/-\alpha$, African) carriers do not produce hydrops/HbH offspring.

5. 1 gene — silent carrier

WHAT YOU SEE

3 genes remain, silent carrier, all gauges normal

WHAT IT ENCODES

Deletion of a single α gene ($-\alpha/aa$) is the silent carrier state: completely normal CBC, normal MCV, normal electrophoresis, no anemia. It is detectable only by DNA analysis and is clinically irrelevant except for genetic counseling. This is the bottom rung of the ladder — the mildest end of the α -thalassemia spectrum.

BOARD TRAP

WRONG: 'A mild microcytic anemia with normal HbA2 must be one-gene silent carrier.' Microcytosis requires loss of TWO genes (trait); the one-gene silent carrier has entirely normal indices.

6. HbC (β_6 Glu $\rightarrow$ Lys)

WHAT YOU SEE

Case 1 orange, crystal intra-RBC, β_6 Glu $\rightarrow$ Lys

WHAT IT ENCODES

HbC results from a β_6 Glutamate $\rightarrow$ LYSINE substitution (same codon as sickle, different amino acid). Homozygous HbCC causes a mild chronic hemolytic anemia with abundant target cells and pathognomonic intracellular HbC crystals (and microspherocytes). Heterozygous HbC trait is asymptomatic. Compound HbSC disease (one S + one C gene) is a MODERATE sickling disorder — milder than HbSS but with more retinopathy, avascular necrosis, and renal papillary necrosis.

BOARD TRAP

WRONG: 'HbC is a Glu $\rightarrow$ Val substitution at β_6 .' Glu $\rightarrow$ VAL at β_6 is HbS (sickle); HbC is Glu $\rightarrow$ LYSINE at the same position — the discriminator is the substituted amino acid (Val vs Lys).

7. HbE (β_{26} Glu $\rightarrow$ Lys)

WHAT YOU SEE

Case 2 yellow, splice site, β_{26} Glu $\rightarrow$ Lys, SE Asia

WHAT IT ENCODES

HbE is a β_{26} Glu $\rightarrow$ Lys mutation that also activates a cryptic splice site, reducing β -chain output (so it behaves partly like a mild β -thalassemia). Homozygous HbEE causes mild microcytosis with target cells and minimal/no anemia. It is the most prevalent Hb variant in Southeast Asia. The board catch: HbE/ β -thalassemia (one HbE + one β -thal allele) can be SEVERE — a major transfusion-dependent thalassemia syndrome.

BOARD TRAP

WRONG: 'HbE is always a benign trait.' HbEE is mild, but co-inheritance as HbE/ β -thalassemia produces a clinically severe, often transfusion-dependent thalassemia.

8. High-O₂-affinity Hb

WHAT YOU SEE

Case 3 blue, ↓P50, won't let go, erythrocytosis

WHAT IT ENCODES

High-affinity hemoglobin variants have a LEFT-shifted O₂ dissociation curve (low P50): they grip O₂ and won't release it to tissues. Tissue hypoxia drives appropriate (secondary) erythrocytosis via EPO — but here SaO₂ is normal and the polycythemia is COMPENSATORY. Diagnose with a low measured P50 and normal/high EPO; there is no JAK2 mutation and no splenomegaly. Usually no treatment is needed.

BOARD TRAP

WRONG: 'High-affinity Hb erythrocytosis should be treated like polycythemia vera with phlebotomy.' This erythrocytosis is compensatory for tissue hypoxia; phlebotomy worsens O₂ delivery. There is no JAK2 mutation — distinguish via low P50 and absent splenomegaly.

9. Low-O₂-affinity Hb

WHAT YOU SEE

Case 4 purple, ↑P50, gives O₂ too fast, cyanosis, mild anemia

WHAT IT ENCODES

Low-affinity hemoglobin variants have a RIGHT-shifted curve (high P50): they release O₂ readily, so venous and tissue O₂ are high and adequate. Because more deoxyhemoglobin is present, patients can appear cyanotic despite a NORMAL PaO₂, and EPO drive is low, giving a mild anemia. The condition is generally benign and requires no therapy — the cyanosis is a red herring.

BOARD TRAP

WRONG: 'Cyanosis with a normal PaO₂ in a well patient means methemoglobinemia.' Low-affinity (high-P50) Hb also causes cyanosis with normal PaO₂ — but blood is not chocolate-brown and co-oximetry is normal; tissue oxygenation is actually good.

10. Unstable Hb

WHAT YOU SEE

Case 5 green, Heinz bodies, heat/isopropanol test

WHAT IT ENCODES

Unstable hemoglobins (e.g., Hb Köln, Hb Hasharon, Hb Zürich) carry mutations that destabilize the globin fold, so the Hb denatures and precipitates as Heinz bodies, producing a Heinz-body hemolytic anemia. Hemolysis is worsened by oxidant stress, drugs, and febrile illness. Diagnose with the heat-stability (50°C) test or isopropanol precipitation test; supravital staining shows Heinz bodies. Management is supportive — avoid oxidant triggers; splenectomy in severe cases.

BOARD TRAP

WRONG: 'Heinz bodies on a smear automatically mean G6PD deficiency.' Heinz bodies also appear in unstable Hb variants (and HbH); the discriminating test is heat/isopropanol stability for unstable Hb vs G6PD enzyme assay.

11. HbM / methemoglobinemia

WHAT YOU SEE

Case 6 grey, Fe³⁺ locked, congenital cyanosis

WHAT IT ENCODES

Hb M denotes globin-chain mutations (usually in the heme pocket, often a proximal/distal histidine substitution) that stabilize iron in the ferric Fe³⁺ state, producing CONGENITAL methemoglobinemia. Patients have lifelong cyanosis that does NOT respond to supplemental O₂ and is generally well tolerated. Because the defect is in globin (not the reductase enzyme), methylene blue does NOT work and no treatment is needed. Inheritance is autosomal dominant.

BOARD TRAP

WRONG: 'Hb M methemoglobinemia is reversed with methylene blue.' Methylene blue works only for enzymatic (cytochrome b5 reductase / acquired oxidant) MetHb; Hb M is a structural globin defect — methylene blue is ineffective and the condition needs no treatment.